

Fuelling a Mining Giant

A diesel, haulage and SARS-refund data story — 23.7 million rows from a mining fleet-operations ERP, modelled as a SQL Server star schema, shipped through Azure Data Factory, and served to Qlik Sense.

290.6M L

fuel issued 2009–2022

325 504

AFS fuel transactions

778 254

equipment trips

R 385.2m

modelled diesel refunds

23.7M

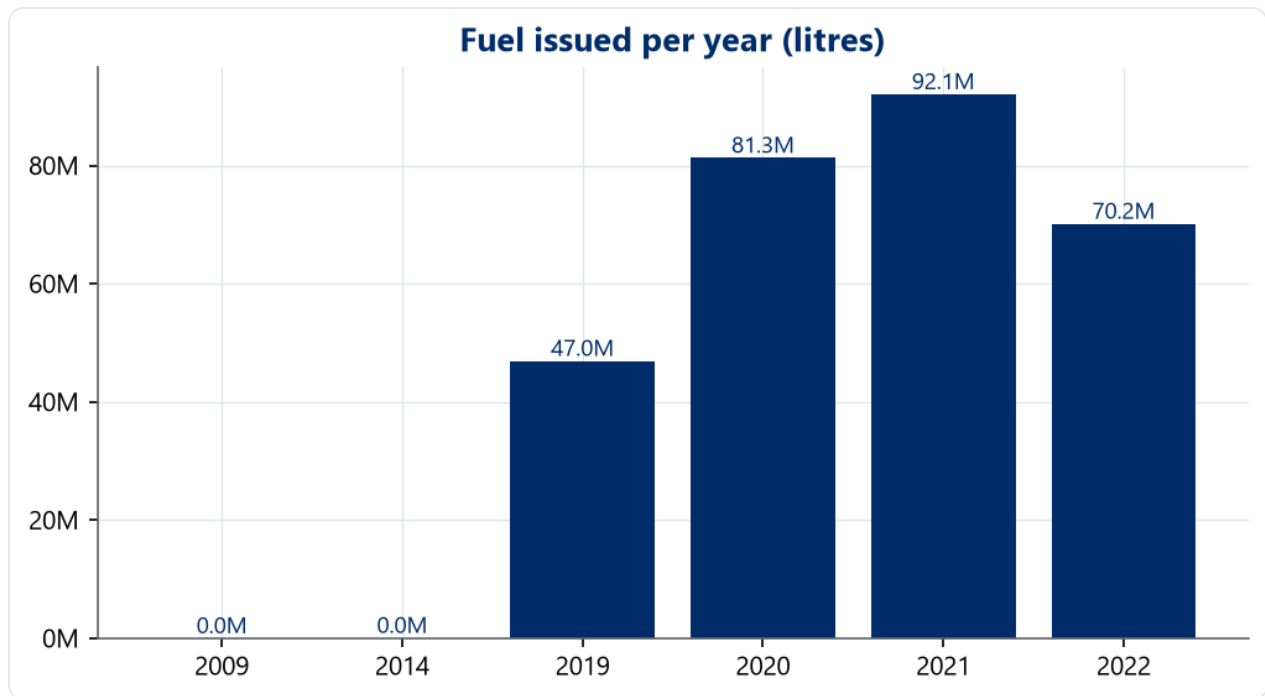
rows through the pipeline

1 • More than fuel records

The source system is an operational ERP for a mining fleet: an Automated Fuel System (AFS) issuing diesel to trucks and drill rigs, tank deliveries, odometer and hour-meter readings, haulage trips with material types, GPS trip traces over a 21.9-million-point coordinate grid, geofences, SAP integration staging and cost-centre accounting. Fuel is the connective tissue — every litre issued ties equipment, location, activity and money together — but the dataset describes how a mine *moves*.

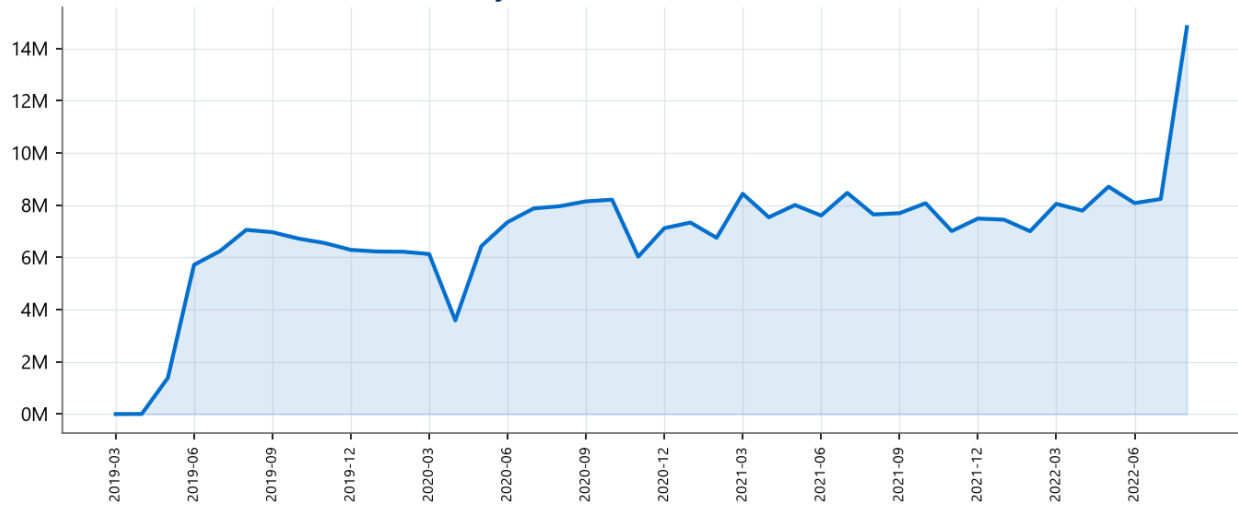
This project models that data as a Kimball star schema (8 dimensions, 8 facts) with one high-stakes business calculation at its centre: the **SARS diesel refund** under Rebate Item 670.04 of the Customs & Excise Act, where classifying litres as eligible or non-eligible is worth millions of rand per month.

2 · The fuel story

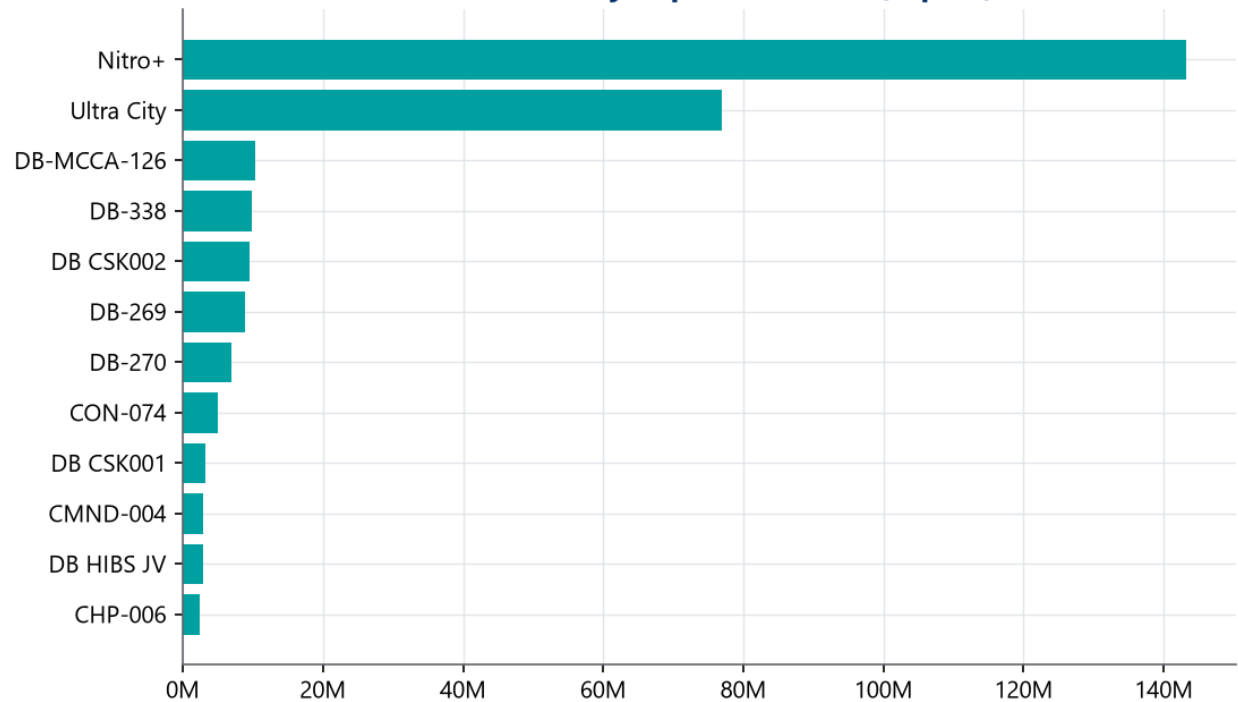


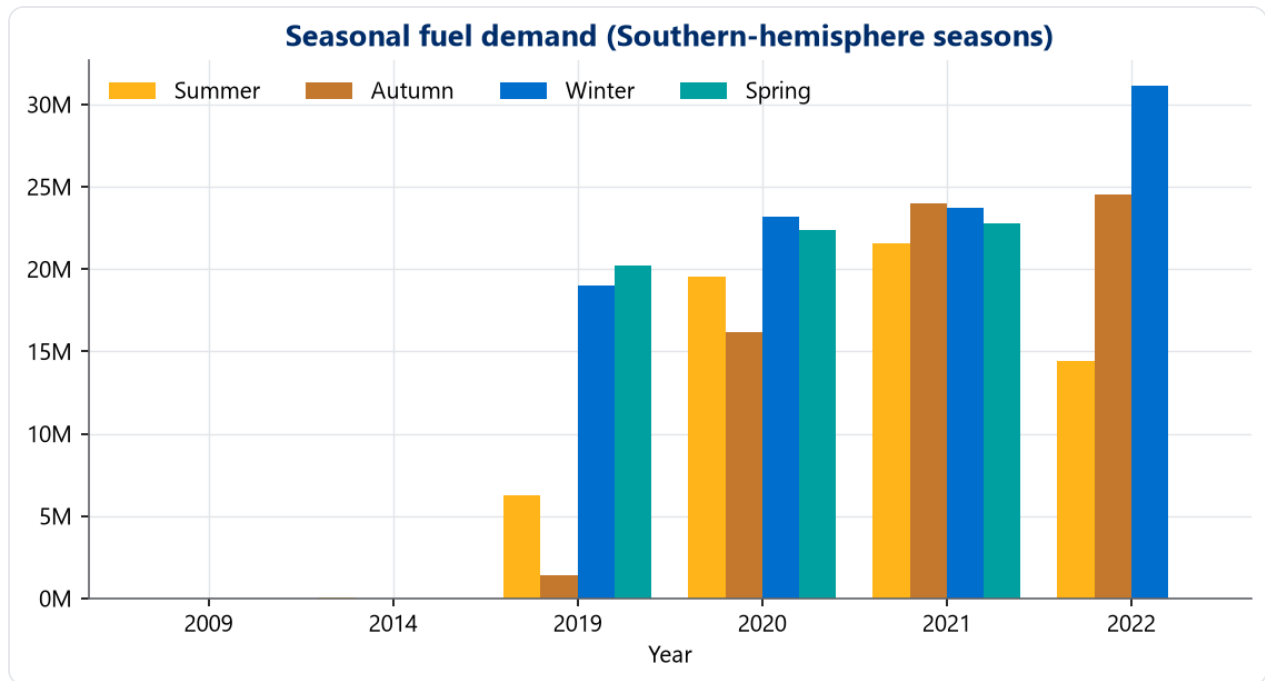
Fuel issue volumes step up sharply from 2019 as the AFS rollout reaches full coverage, settling around 70–80 million litres a year across the operation.

Monthly fuel issued (litres, 2019 onward)



Fuel issued by depot / location (top 12)

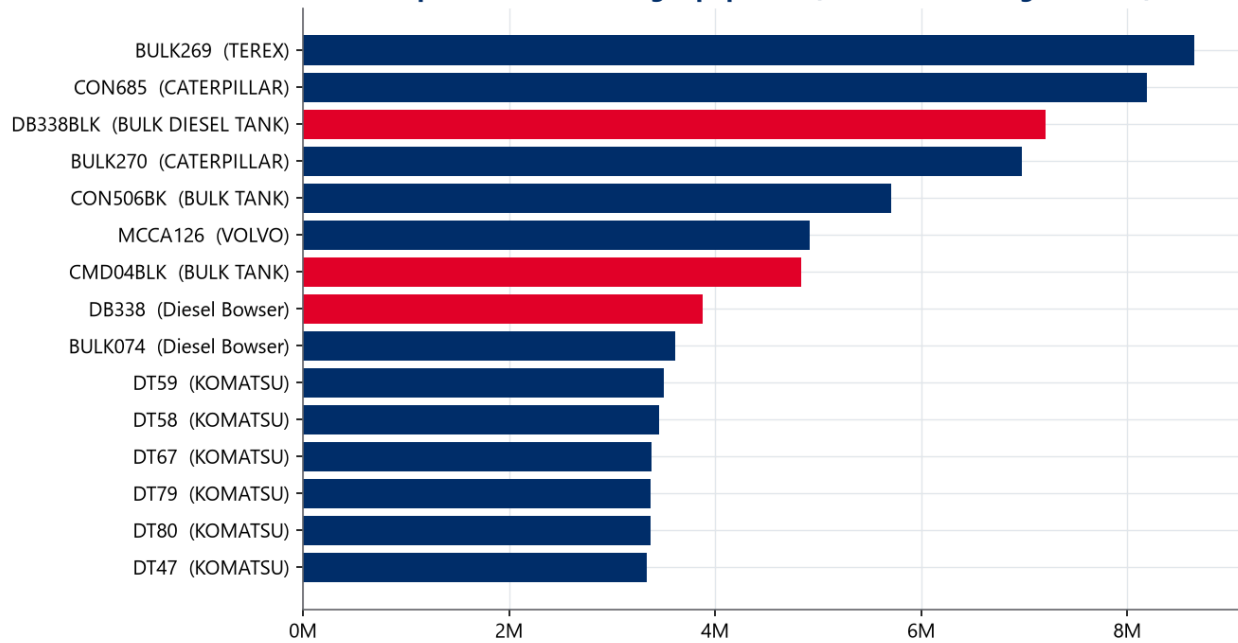




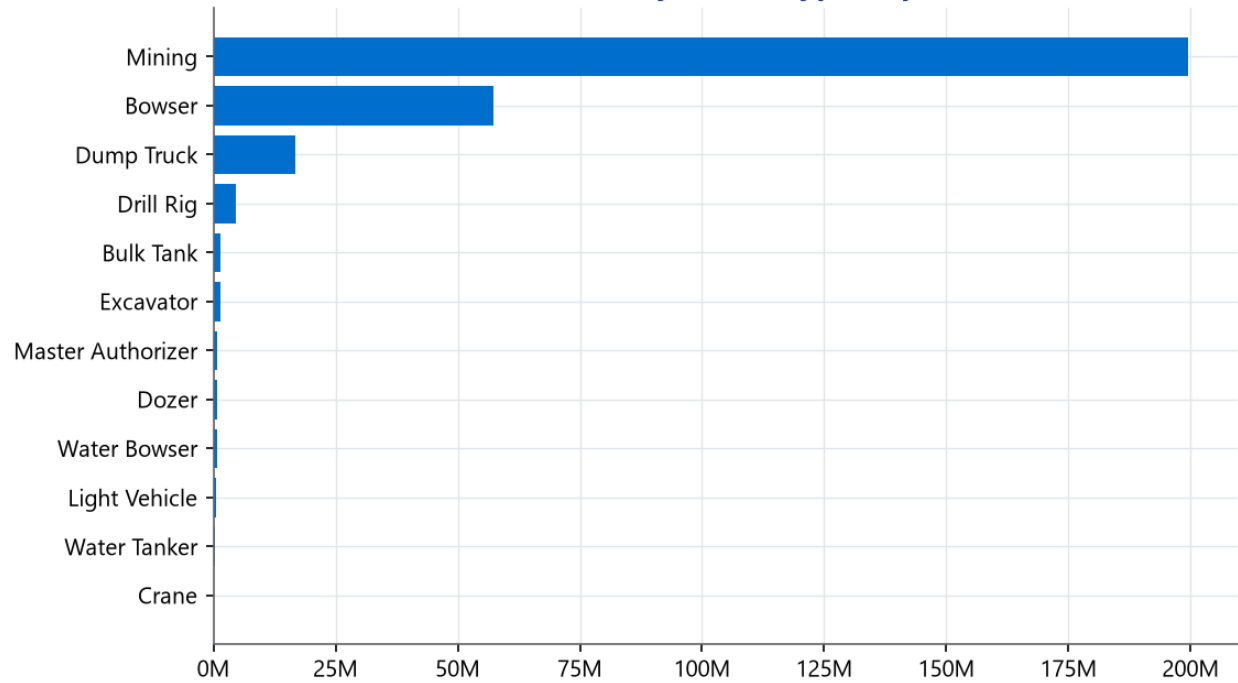
Demand is not smooth: maintenance shutdowns, production cycles and season shape the curve — one reason a refund forecast needs more than a flat average.

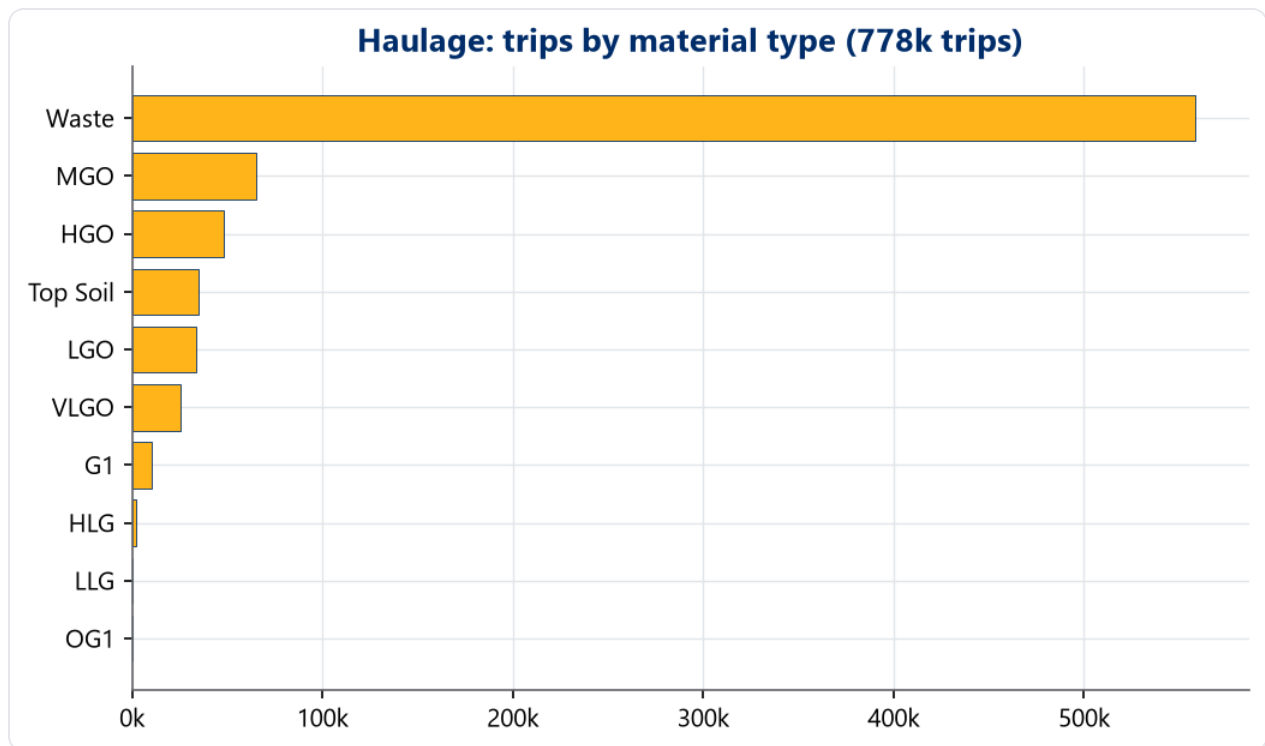
3 · Fleet & haulage

Top 15 fuel-consuming equipment (red = SARS-ineligible class)

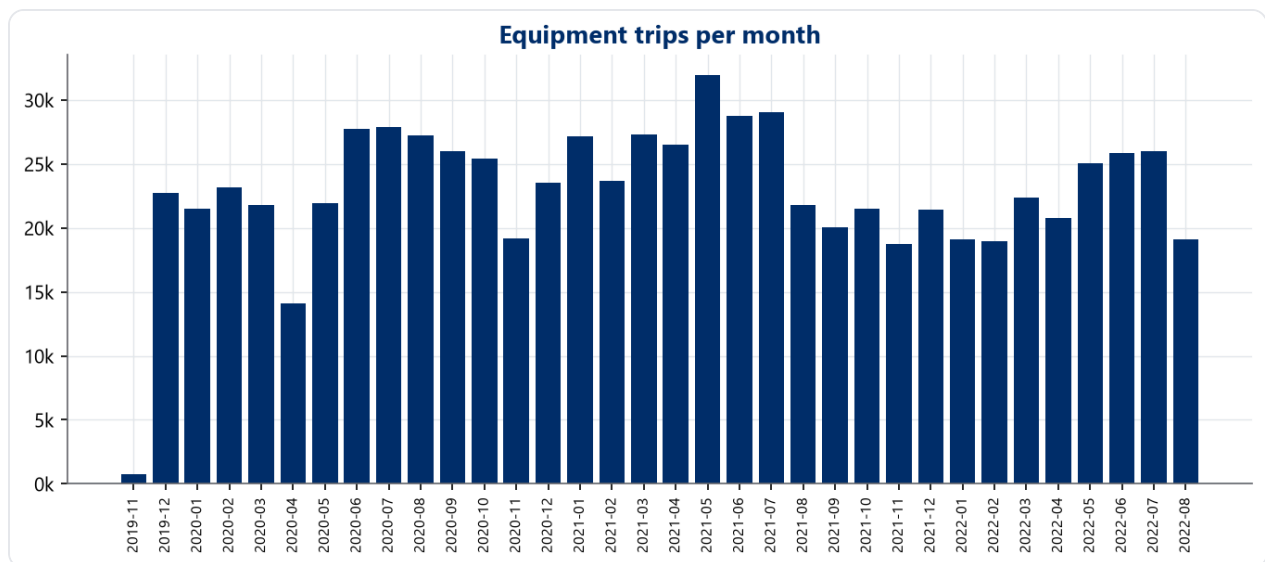


Fuel issued by vehicle type (top 12)





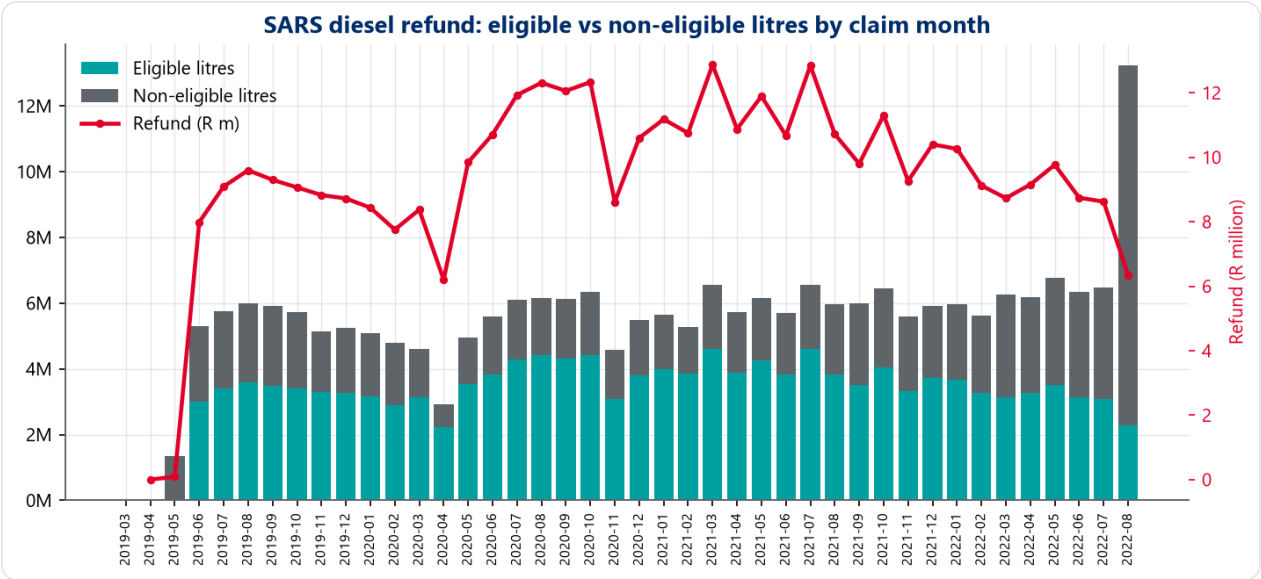
Trip records carry the material being moved — ore grades, waste, consumables — which is exactly the evidence SARS eligibility classification leans on: transport of ore on a mining site is claimable, general road use is not.



4 · The SARS diesel refund

For on-land primary producers (mining included), the refund formula from the policy evidence is:

```
eligible_litres    = total_litres - non_eligible_litres
qualifying_litres = eligible_litres × 80%
refund_rand       = qualifying_litres × refund_rate (c/L) ÷ 100
```



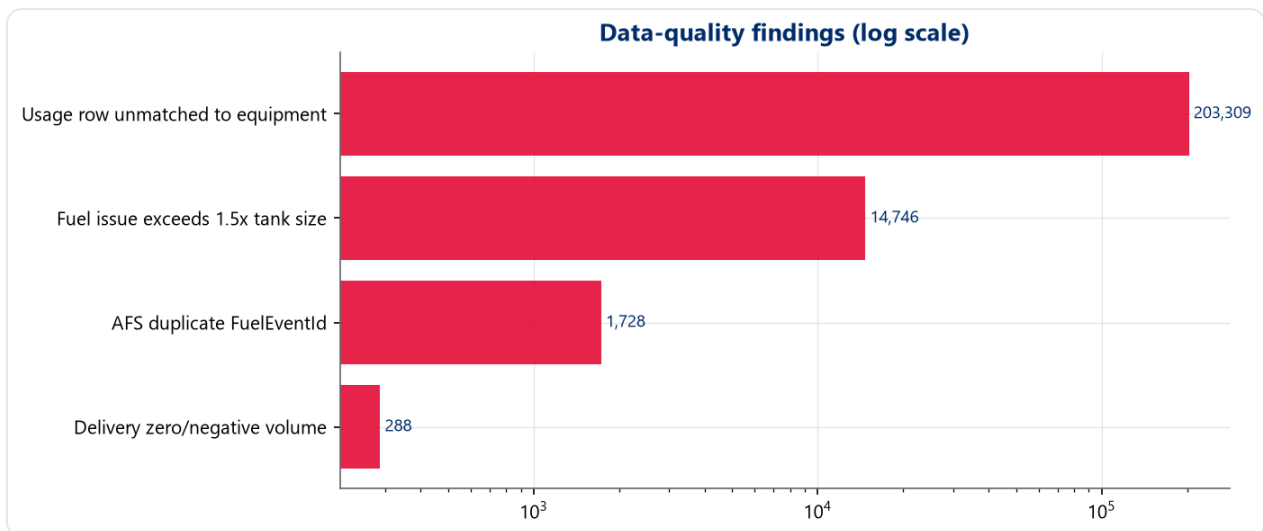
Last 12 claim months

Claim month	Total L	Non-eligible L	Eligible L	Qualifying L	Refund
2021-09	5 982 077	2 471 924	3 510 153	2 808 122	R 9 800 347.18
2021-10	6 455 776	2 405 818	4 049 958	3 239 966	R 11 307 482.74
2021-11	5 604 740	2 285 870	3 318 870	2 655 096	R 9 266 285.04
2021-12	5 922 270	2 197 315	3 724 955	2 979 964	R 10 400 075.48
2022-01	5 972 874	2 297 767	3 675 107	2 940 086	R 10 260 899.86
2022-02	5 612 826	2 345 327	3 267 500	2 614 000	R 9 122 859.44
2022-03	6 272 003	3 143 907	3 128 096	2 502 477	R 8 733 643.19
2022-04	6 193 891	2 914 930	3 278 961	2 623 169	R 9 154 858.83
2022-05	6 778 019	3 279 961	3 498 059	2 798 447	R 9 766 579.61
2022-06	6 344 588	3 212 236	3 132 352	2 505 881	R 8 745 525.95

2022-07	6 473 722	3 384 942	3 088 781	2 471 025	R 8 623 875.99
2022-08	13 224 056	10 952 680	2 271 376	1 817 101	R 6 341 680.95

Caution: rates are the 2020 SARS policy examples found in the project's evidence pack (349 c/L on-land after 1 Apr 2020). They are implementation evidence, not tax advice — update `dw.DimRefundRate` with current rates before relying on any figure.

5 · Data quality — where the money leaks



Finding	Rows
Usage row unmatched to equipment	203 309
Fuel issue exceeds 1.5x tank size	14 746
AFS duplicate FuelEventId	1 728
Delivery zero/negative volume	288

The stand-outs: over 200 000 usage-logbook rows carry registration numbers that don't match the equipment master (an audit-trail gap for refund claims), and ~14 700 single fuel issues exceed 1.5× the receiving vehicle's tank size — classic candidates for meter faults or leakage/theft review. One equipment-master row even had a tab character embedded in its

registration number, found because it was the only row in 23.7 million that broke a rectangular file export.

6 · How it's built

```
SQL Server (AngloData QA)           — local, always works offline
└─ dw star schema · 8 dims, 8 facts, 5 views (01_create_load_dw_full.sql)
    └─ TSV export (bcp, UTF-8)          (02_export_dw_to_tsv.ps1)
        └─ curated_local/ Parquet      (03_build_local_parquet.py)
            └─ azcopy → ADLS Gen2 raw/  — Azure mirror
                └─ Data Factory pl_raw_to_curated (ForEach Copy, TSV → snappy Parquet)
                    └─ curated/dw/<Table>/*.parquet
                        └─ Qlik Sense Cloud (SAS web-files or Azure Storage connector)
```

The warehouse reconciles exactly: 290 557 288.29 litres issued in both source and fact table, to the hundredth of a litre. Everything regenerates from five scripts, and the Azure resource group tears down with one command when it has served its purpose.